

Stochastic Modeling of Bovine Respiratory Disease Using Agent-Based Modeling

Evan Kniffen • Department of Animal Science, Texas A&M University

Introduction

Goal. Provide a mathematical understanding of a NetLogo agent-based model (ABM) that quantifies BRD impacts across the cow–calf (CC), stocker (ST), and feedlot (FL) pipeline, and benchmark it against a mean-field SEIQRS model.

Outputs. Weight/ADG distributions, BRD incidence/timing, antimicrobial use (mg/PCU), and economics (NPV).

Validation. ABM ensemble vs. SEIQRS ODE model: peak time/height, final size, and $R_e(t)$ trajectory.

Why ABM? Captures heterogeneity (phase, weight), stochastic infection clusters, and treatment rules; math ensures transparency (identifiable (A, b, k) and rate-consistent hazards).

Model Scope

Per-agent state at day t :

$$H_t \in \{S, E, I, R\}, \quad W_t \in \mathbb{R}_+, \quad \phi_t \in \{\text{CC, ST, FL}\}, \quad V_t = p_{\phi_t} W_t.$$

Population counts: $N_t = S_t + E_t + I_t + R_t$.

Objective: Build a discrete-time marked process $\{(H_t, W_t, \phi_t, V_t)\}_{t \geq 0}$ whose drift/noise are interpretable, estimable, and rate-consistent.

Why. Day-scale management and BRD dynamics justify $\Delta t = 1$ d ABM state; microstates enable heterogeneity and pathwise economics.

Gompertz Growth

Closed form (phase ϕ): $W_\phi(t) = A_\phi e^{-b_\phi e^{-k_\phi t}}$. **ODE:** $W' = k_\phi W \ln(A_\phi/W)$.

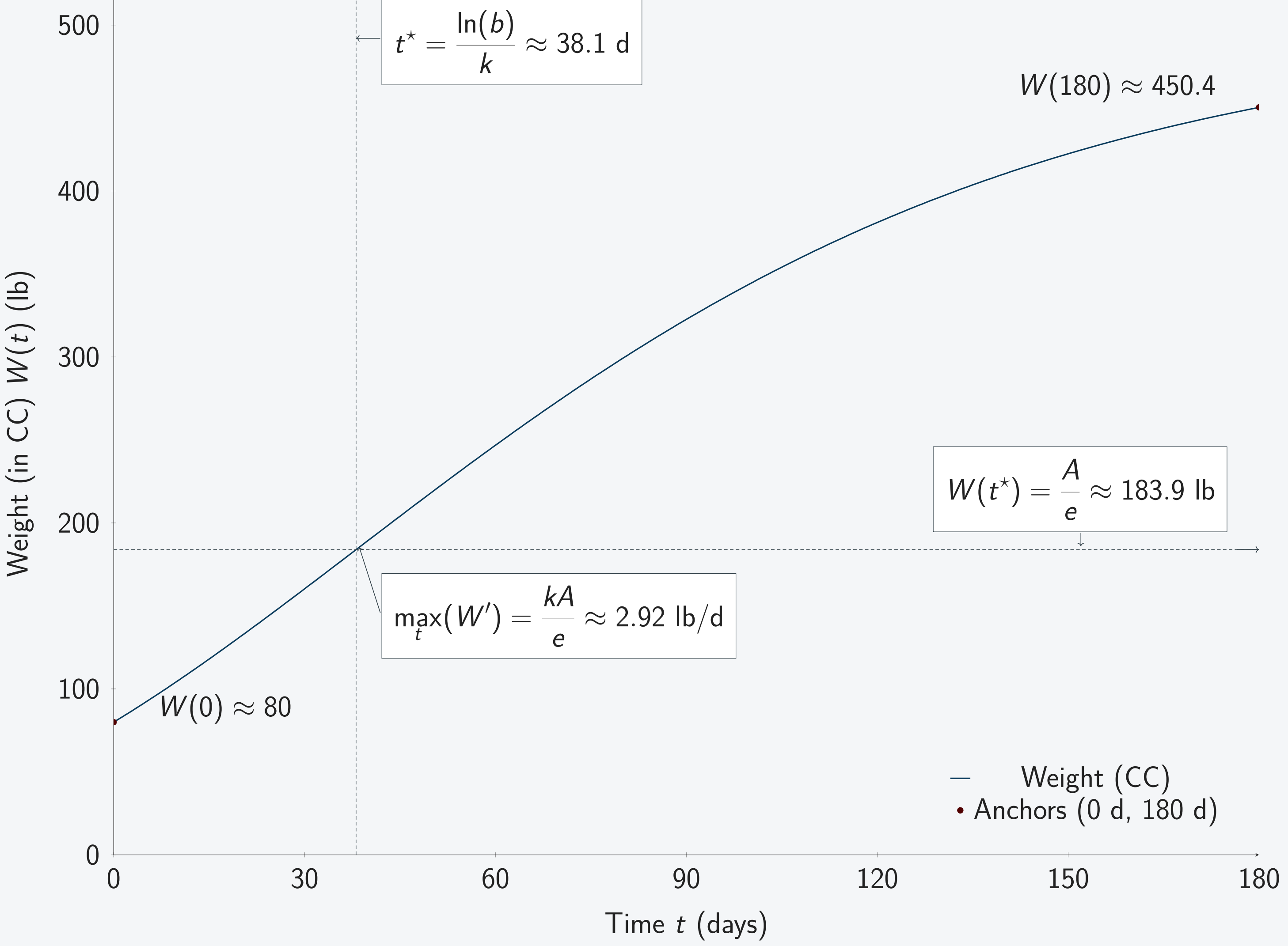
Linearization: $Y(t) = \ln[-\ln(W/A_\phi)] = \ln b_\phi - k_\phi t$.

Inflection: $t^* = \frac{1}{k_\phi} \ln b_\phi$, $W(t^*) = A_\phi/e$, $\max_t(W') = \frac{k_\phi A_\phi}{e}$.

Two-point ID (given A_ϕ):

$$k_\phi = \frac{\ln[-\ln(W_1/A_\phi)] - \ln[-\ln(W_2/A_\phi)]}{t_2 - t_1}, \quad b_\phi = -\ln(W_1/A_\phi) e^{k_\phi t_1}.$$

Gompertz Curve during CC:



Phase-Specific Parameterization

Choose A_ϕ and reference targets $(t_0, W_0), (t_0+180, W_{180})$ per phase. Then compute (b_ϕ, k_ϕ) by the formula above and carry them into the stochastic model. Typical choices (illustrative):

Phase	W_0	W_{180}	A_ϕ	(b_ϕ, k_ϕ)
CC	80	450	500	(1.833, 0.0159)
ST	450	800	900	(0.693, 0.00984)
FL	800	1125	1400	(0.5596, 0.00522)

Note: Values shown are from a two-point calibration with $\Delta t = 180$ d.

Contact: kniff@tamu.edu

Discrete Stochastic Growth

Stochastic Daily Weight Gain:

$$W_{t+1} = W_t + \gamma_t (W_\phi^*(t) - W_t) + \varepsilon_t, \quad W_\phi^*(t) = A_\phi e^{-b_\phi e^{-k_\phi t}}, \quad \varepsilon_t \sim \mathcal{N}\left(0, \sigma^2(\alpha + \beta W_\phi^*(t))^2\right).$$

Mean recursion: with $m_t := \mathbb{E}[W_t]$,

$$m_{t+1} = (1 - \gamma_t) m_t + \gamma_t W_\phi^*(t).$$

If $\gamma_t \equiv \gamma$ and $W_\phi^*(t) \equiv W^*$ locally, then

$$m_t = W^* + (1 - \gamma)^{t-t_0} (m_{t_0} - W^*) \quad (\text{geometric mean-reversion}).$$

General step size Δt using Euler–Maruyama:

$$W_{t+\Delta t} = W_t + \kappa_t \Delta t (W_\phi^*(t) - W_t) + \sigma(\alpha + \beta W_\phi^*(t)) \sqrt{\Delta t} \xi_t, \quad \xi_t \sim \mathcal{N}(0, 1).$$

For $\Delta t = 1$ day, set $\gamma_t \equiv \kappa_t$ and $\varepsilon_t = \sigma(\alpha + \beta W_\phi^*(t)) \xi_t$ (the discrete model above).

SDE analogue:

$$dW_t = \kappa_t (W_\phi^*(t) - W_t) dt + \sigma(\alpha + \beta W_\phi^*(t)) dB_t.$$

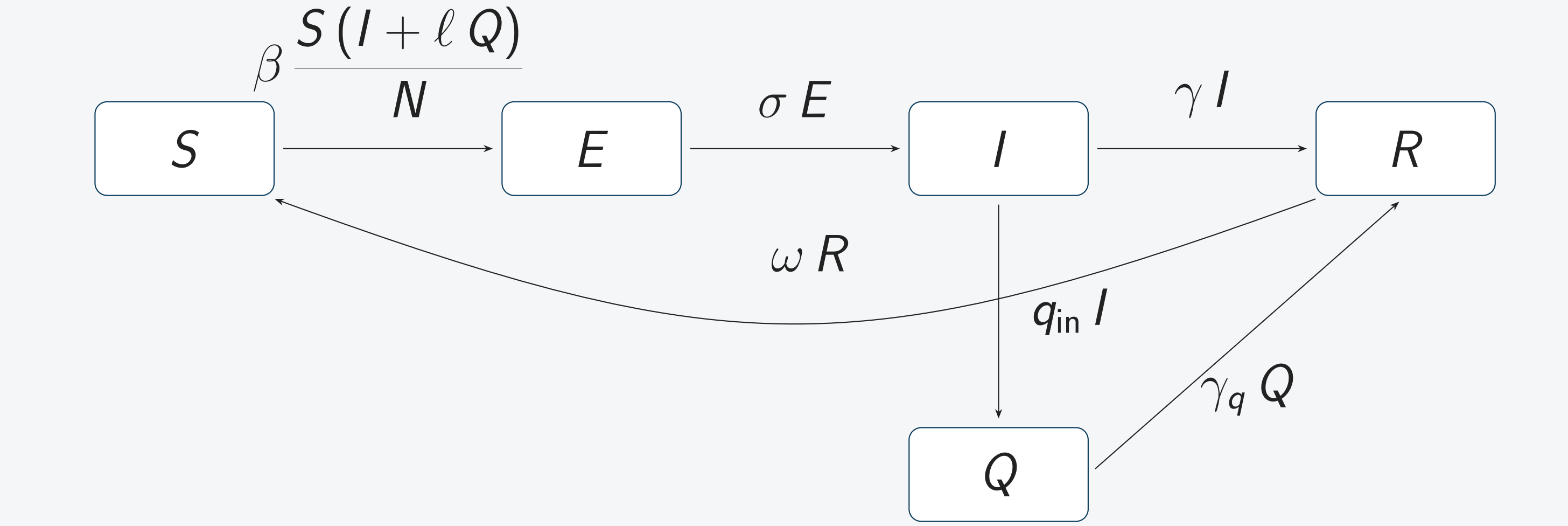
With bounded κ_t and $0 < W_\phi^*(t) \leq A_\phi$, drift and diffusion coefficients satisfy linear-growth bounds and global Lipschitz conditions $\Rightarrow$ strong solution exists and is unique.

Infection suppression (drift scaling):

$$\gamma_t = \exp(-\delta_1 \mathbf{1}_{\{\text{acute}\}} - \delta_2 \mathbf{1}_{\{\text{prolonged}\}}),$$

which slows mean reversion while preserving the same $W_\phi^*(t)$ target and non-negativity.

Epidemiology



Deterministic Model. With $S, E, I, Q, R \in [0, 1]$ and $S+E+I+Q+R \equiv 1$:

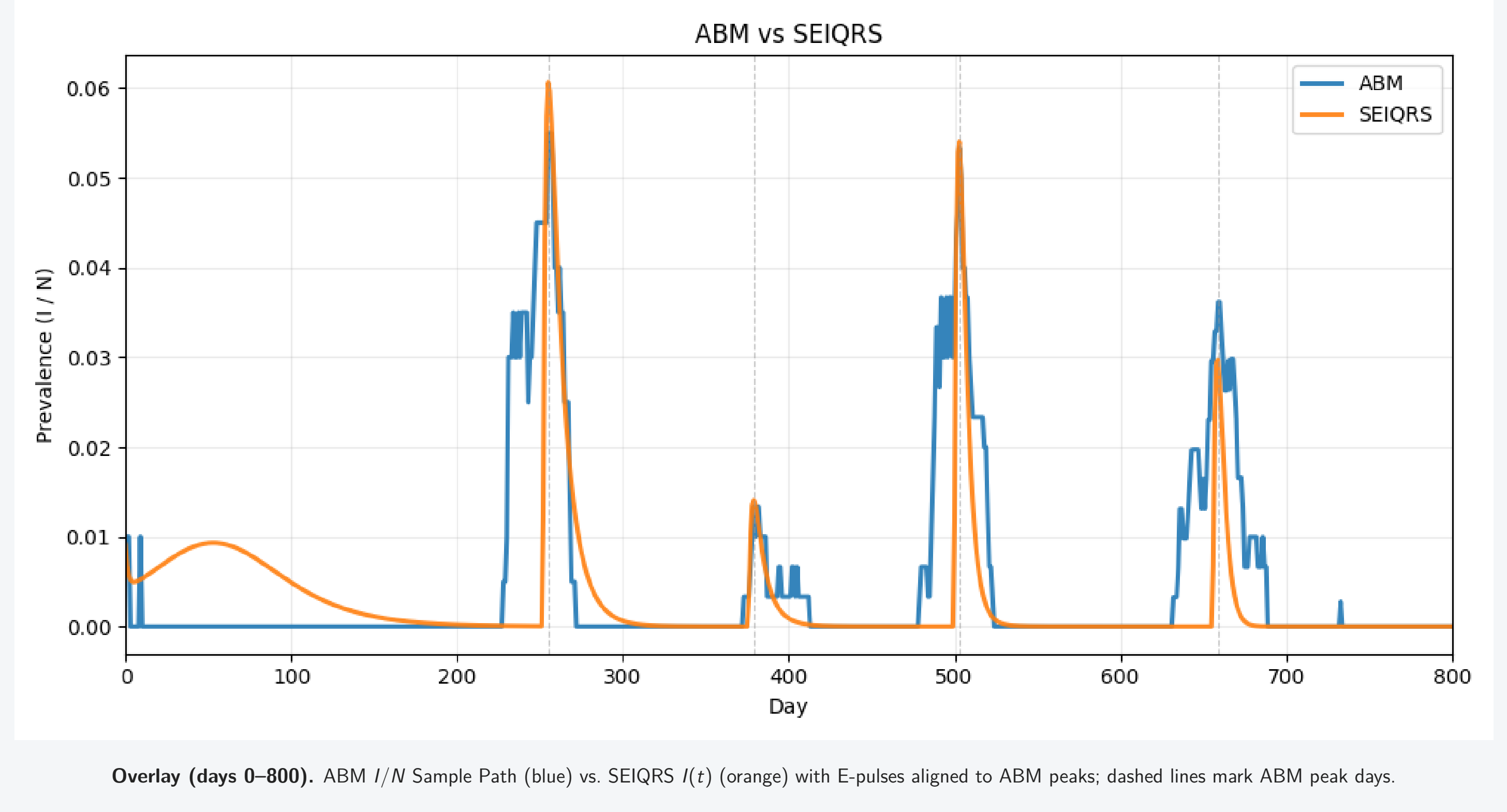
$$\begin{aligned} S' &= -\beta S(I + \ell Q) + \omega R, \\ E' &= \beta S(I + \ell Q) - \sigma E + \sum_k a_k \delta(t - \tau_k), \\ I' &= \sigma E - (\gamma + q_{\text{in}})I + q_{\text{out}}Q, \\ Q' &= q_{\text{in}}I - (\gamma_q + q_{\text{out}})Q, \\ R' &= \gamma I + \gamma_q Q - \omega R. \end{aligned}$$

$\ell \in [0, 1]$ is the quarantine leak (reduced infectiousness in Q); q_{in} moves $I \rightarrow Q$, q_{out} returns $Q \rightarrow I$ (often 0), and γ_q recovers in Q . Daily ABM hazards map directly to rates $(\beta, \sigma, \gamma, \omega)$ $[\text{d}^{-1}]$.

Impulse alignment. Treat each ABM infection as an E –pulse at $\tau_k = \hat{t}_k - \text{lag}$ and apply

$$E(\tau_k^+) = E(\tau_k^-) + a_k, \quad S(\tau_k^+) = S(\tau_k^-) - a_k,$$

so $S+E+I+Q+R$ stays 1. Choose $a_k \approx \min(p_k/g_{\text{max}}, 0.3)$, where p_k is the ABM peak height and $g_{\text{max}} = \max_t I(t)$ after a unit E pulse (computed once with $S \approx 1$).



Overlay (days 0–800). ABM I/N Sample Path (blue) vs. SEIQRS $I(t)$ (orange) with E-pulses aligned to ABM peaks; dashed lines mark ABM peak days.

Hazards

Hazard function. For a hazard $\lambda(t)$ on $[t, t + \Delta t]$, an event has a survival S and event probability p given by

$$S(\Delta t) = \exp\left(-\int_t^{t+\Delta t} \lambda(u) du\right), \quad p(t) = 1 - S(\Delta t).$$

If λ is held constant on the step, $p = 1 - e^{-\lambda \Delta t} = \lambda \Delta t - \frac{1}{2}(\lambda \Delta t)^2 + \mathcal{O}((\lambda \Delta t)^3)$.

CTMC generator Q for SEIQRS. Let the state order be (S, E, I, Q, R) and write $\Lambda_{SE}(t) = \beta \frac{I(t) + \ell Q(t)}{N(t)}$.

$$Q = \begin{pmatrix} -\Lambda_{SE} & \Lambda_{SE} & 0 & 0 & 0 \\ 0 & -\sigma & \sigma & 0 & 0 \\ 0 & 0 & -(\gamma + q_{\text{in}}) & q_{\text{in}} & \gamma \\ 0 & 0 & q_{\text{out}} & -(\gamma_q + q_{\text{out}}) & \gamma_q \\ \omega & 0 & 0 & 0 & -\omega \end{pmatrix}, \quad P(\Delta t) = e^{Q \Delta t}.$$

Rows sum to 0; entries are per-capita transition rates between compartmental states.

Exact per-step transition probabilities (size Δt). For single-outflow states:

$$p_{SE}(t) = 1 - \exp\left(-\beta \frac{I_t + \ell Q_t}{N_t} \Delta t\right), \quad p_{EI}(t) = 1 - e^{-\sigma \Delta t}, \quad p_{RS}(t) = 1 - e^{-\omega \Delta t}.$$

For competing-risk states, define $\Lambda_I = \gamma + q_{\text{in}}$ and $\Lambda_Q = \gamma_q + q_{\text{out}}$:

$$\begin{aligned} \Pr(I \rightarrow R) &= (1 - e^{-\Lambda_I \Delta t}) \frac{\gamma}{\Lambda_I}, & \Pr(I \rightarrow Q) &= (1 - e^{-\Lambda_I \Delta t}) \frac{q_{\text{in}}}{\Lambda_I}, \\ \Pr(Q \rightarrow R) &= (1 - e^{-\Lambda_Q \Delta t}) \frac{\gamma_q}{\Lambda_Q}, & \Pr(Q \rightarrow I) &= (1 - e^{-\Lambda_Q \Delta t}) \frac{q_{\text{out}}}{\Lambda_Q}. \end{aligned}$$

In each case, $\Pr(\text{no event}) = e^{-\Lambda \Delta t}$ with $\Lambda \in \{\Lambda_I, \Lambda_Q\}$.

Time-varying hazards. With $\lambda(u)$ changing within the day (e.g.

$$\Lambda_{SE}(u) = \beta [I(u) + \ell Q(u)]/N(u),$$

$$p = 1 - \exp\left(-\int_t^{t+\Delta t} \lambda(u) du\right) \approx 1 - \exp\left(-\lambda\left(t + \frac{\Delta t}{2}\right) \Delta t\right).$$

The explicit Euler choice $\lambda(t)$ gives the simpler $p \approx 1 - e^{-\lambda(t) \Delta t}$ used in our ABM.

Population draws. For single-outflow states:

$$X_{SE} \sim \text{Binomial}(S_t, p_{SE}(t)), \quad X_{EI} \sim \text{Binomial}(E_t, p_{EI}(t)), \quad X_{RS} \sim \text{Binomial}(R_t, p_{RS}(t)).$$

NetLogo ABM Implementation

Setup.

- Build animal agents with tuples (H_0, W_0, ϕ_0) and unique IDs; start infection/treatment/phase timers.
- Tag pens/patches by phase (CC, ST, FL) and a dedicated *quarantine/sick-bay*; set capacities and movement schedule.
- Load parameters (growth, disease, detection/treatment, prices/costs, withdrawal time) and initialize logs/monitors.

Go function modules (run daily).

1. *Population* — maintain headcount; advance ages/weights; handle scheduled batch moves (wean, ship).
2. *Movement & Integration* — route batches CC→ST→FL; assign pens on arrival; capture stochastic movement of each turtle.
3. *Vaccination & Immunity* — apply scheduled vaccines/boosters; track protection/waning and coverage/compliance.
4. *Contact & Transmission* — compute within-pen mixing with environmental contamination/decay and update H_t states.
5. *Quarantine* — move detected sick animals to isolation; reduce contacts & weight; adjust performance/costs until release.
6. *Treatment & AMU Decisions* — run metaphylaxis/therapy rules, apply dosing and withdrawal clocks; tally AMU metrics.

Outputs.

- Epidemiology: S_t, E_t, I_t, R_t , incidence, durations in H_t .
- AMU: mg/PCU (daily/cumulative), courses/1,000 animal-days, withdrawal compliance.
- Operations & economics: movement tracebacks, value V_t , costs, scenario summaries (NPV, median/IQR).

Overall, ABMs shine because they model individual animals, contacts, and policy directly. They capture heterogeneity, stochastic outbreaks, and constraints that ODEs can miss.